F24-069-D-VocaLink

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Abstract:

Vocalink is an AI-driven voice-native text editor aimed at improving accessibility and efficiency for users, particularly those with neurodivergent conditions like Attention Deficit Hyperactivity Disorder (ADHD), dyslexia, and dysgraphia. Traditional dictation systems often lack the flexibility and accuracy needed for hands-free text creation, imposing cognitive and motor challenges. *Vocalink* overcomes these limitations by employing advanced speech recognition technologies, such as Mixture of Experts (MoE) models, near-real-time contextual processing, and natural language understanding. This approach allows for accurate, command-free document creation, making it easier for users to format, edit, and manage documents through voice commands.

Executive Summary:

Vocalink is a breakthrough AI-powered, voice-native text editor designed to offer a more inclusive and accessible way for people to create documents. It addresses the limitations of existing dictation systems, which often struggle with accuracy, rigid commands, and the need for significant post-editing. This is especially challenging for individuals with neurodivergent conditions like ADHD, dyslexia, and dysgraphia, as well as professionals who require efficient, hands-free tools for their work.

Problem Statement:

Current voice dictation technologies, like *Google Voice Typing* and *Apple Dictation*, often fail to meet the needs of users who require greater flexibility and accuracy in their workflows. These tools typically rely on rigid command structures and require manual corrections, making them difficult to use, particularly for those who face cognitive or motor coordination challenges.

In a world where more people with neurodivergent conditions are entering academic and professional settings, there is a growing need for tools that reduce the cognitive load associated with document creation. *Vocalink* meets this need by offering an intuitive, hands-free solution that minimises the frustrations of traditional dictation tools.

Proposed Solution:

Vocalink stands out from existing tools by leveraging advanced AI technologies to provide a smoother, more efficient document creation process. Using the Mixture of Experts (MoE) model, *Vocalink* can dynamically activate specialised AI "experts" based on the input, resulting in more accurate speech recognition and better contextual understanding. This allows users to dictate, edit, and format documents naturally and hands-free, with far fewer errors and manual interventions.

Key features of *Vocalink* include:

- **Near-Real-Time Speech-to-Text Conversion:** Converts spoken words into text with minimal delay, ensuring a seamless dictation experience.
- **Natural Command Execution:** Users can interact with their documents using intuitive voice commands, like "bold this sentence" or "start a new paragraph," without needing to memorise rigid command structures.
- **Adaptive Error Correction:** Automatically corrects common speech recognition mistakes based on context, so users don't need to stop and manually edit.

- **Multilingual Support:** *Vocalink* supports both English and Roman Urdu/Hinglish, enabling users to switch between languages smoothly and naturally.

- **Accessibility for Neurodivergent Users:** Designed specifically to reduce the mental effort involved in document creation, Vocalink's automatic formatting and error correction features make it especially useful for individuals with ADHD, dyslexia, and similar conditions.

Business Opportunity:

The assistive technology market is experiencing rapid growth, driven by the increasing demand for tools that cater to diverse cognitive and accessibility needs. This market is projected to grow from **$22.98 billion** in **2023** to **$32.25 billion** by **2030**. *Vocalink* is well-positioned to capture a segment of this market by targeting not only neurodivergent users but also professionals who need efficient, hands-free tools for document creation.

The voice recognition and dictation market is also expanding rapidly, with the speech-to-text segment alone expected to grow from **$7.14 billion** in **2024** to **$15.87 billion** by **2030**. *Vocalink's* focus on inclusivity, multilingual support, and hands-free efficiency gives it a competitive advantage in this growing market. Even a modest market share could translate into significant revenue, highlighting the business potential of the product.

Mixture of Experts (MoE) Approach:

At the core of *Vocalink's* technology is the Mixture of Experts (MoE) model, an advanced AI framework that ensures high accuracy and efficiency in speech-to-text conversion. Unlike traditional systems that rely on a single algorithm, the MoE approach dynamically activates specialised AI experts based on the user's input. This approach is based on the work of Jiang et al. (2024) in their paper "Mixtral of Experts," which demonstrates how dynamic expert selection can improve performance and optimize resource use.

These experts handle different aspects of speech recognition, such as contextual understanding, natural language processing, and error correction, ensuring that each task is processed by the most appropriate model. This leads to more accurate transcriptions, better understanding of user intent, and faster processing times, all while optimising system resources. The MoE model sets *Vocalink* apart, providing a flexible and highly adaptive voice-to-text solution.

Conclusion:

Vocalink is more than just another voice dictation tool. It's a comprehensive, AI-powered solution that redefines how people create and interact with documents, particularly for those who find traditional dictation systems limiting. By reducing the cognitive load and offering a more flexible, accurate experience, *Vocalink* is poised to make a lasting impact on how people work and communicate.

Chapter 1

Introduction

Vocalink is an AI-driven voice-native text editor designed to address the limitations of traditional dictation systems and support a wide range of users, including those with neurodivergent conditions such as Attention Deficit Hyperactivity Disorder (ADHD), dyslexia, and dysgraphia. Conventional dictation technologies often struggle with accuracy, rigid command structures, and the need for manual post-editing, making them less accessible to individuals with cognitive and motor challenges.

As more people with neurodivergent conditions enter academic and professional settings, the need for inclusive tools that reduce the cognitive load associated with text generation is increasingly critical. *Vocalink* responds to this challenge by combining advanced speech recognition technologies, such as Mixture of Experts (MoE) models, near-real-time contextual processing, and natural language understanding. This enables a more intuitive, hands-free document creation experience that minimises manual corrections and command-driven interactions.

Beyond neurodivergent users, *Vocalink* offers significant benefits for professionals in various fields by enabling hands-free document creation for tasks like drafting meeting notes, project updates, and reports. It enhances accessibility for individuals with physical disabilities or temporary injuries, allowing them to work more efficiently without the need for typing. Its support for Roman Urdu and Hinglish makes it ideal for users who switch between these languages, enhancing productivity and communication.

1.1 Research on Existing Products

In order to better understand the competitive landscape and position *Vocalink* effectively, a comparison was conducted against leading voice recognition tools. The table below highlights key features offered by *Vocalink* alongside other widely used solutions such as *Nuance Dragon*, *Apple Dictation*, *Google Voice Typing*, and *ChatGPT Voice (GPT-4)*.

This analysis provides insight into how *Vocalink* differentiates itself through features like adaptive formatting, flexible language commands, and speaker recognition, which are not universally available in other tools.

Feature	**Vocalink**	**Nuance Dragon**	**Apple Dictation**	**Google Voice Typing**	**ChatGPT Voice (GPT-4o)**
Real-Time Dictation	✓	✓	✓	✓	✓
Adaptive Formatting	✓	✓			
Flexible Language Commands	✓	✓			
Context Awareness	✓	✓	✓	✓	✓
Intent Recognition	✓	✓	✓		✓
Speaker Recognition	✓				
Text-Based Chat	✓				

Table 1.1: Feature Comparison of Vocalink and Leading Voice Recognition Tools

Chapter 2

Project Vision

2.1 Problem Statement

The increasing prevalence of neurodivergent conditions such as dyslexia, Attention Deficit Hyperactivity Disorder (ADHD), and dysgraphia in both academic and professional environments underscores the need for more inclusive tools that cater to diverse cognitive needs. Individuals with these conditions often face challenges with traditional methods of document creation, including typing and voice dictation technologies. Typing demands sustained cognitive and motor coordination, while current dictation technologies lack the contextual awareness, accuracy, and flexibility necessary to fully accommodate these users.

Existing voice dictation systems, such as *Google Voice Typing* and *Apple Dictation*, are characterised by high word error rates and rigid command structures, resulting in the need for significant post-editing. This additional cognitive load undermines the intended benefits of voice dictation, particularly for individuals with conditions like dyslexia or ADHD, who may struggle with attention and task management. Despite advances in voice recognition technology, current systems do not provide an intuitive, seamless experience that reduces manual corrections and reliance on command-driven interactions.

2.2 Proposed Solution

Vocalink is an AI-powered, voice-native text editor designed to address the limitations of traditional dictation systems, such as low accuracy, rigid command structures, and the need for extensive manual editing. By leveraging MoE models, near-real-time processing, and contextual awareness, *Vocalink* reduces WER, enhances speech-to-text accuracy, and enables hands-free, natural document creation.

- **Mixture of Experts (MoE) Architecture:** *Vocalink* utilises MoE models, wherein specialised AI agents are selectively activated based on the characteristics of the input. This leads to more efficient processing, improving speech recognition and reducing WER. The system dynamically allocates appropriate experts to tasks such as phrase structuring and error correction, ensuring high accuracy while optimising computational resources.

- **Near-Real-Time Contextual Processing:** *Vocalink* processes speech with minimal latency, allowing for quick interpretation of user intent and continuous text refinement. It supports natural interactions by distinguishing between content, commands, and filler words. Spoken instructions such as "delete the last sentence" are executed seamlessly without the need for predefined commands, resulting in a fluid and efficient workflow.

- **Adaptive Error Correction:** *Vocalink* automatically corrects common speech errors and adapts content based on context. By recognising and refining errors such as misstatements (e.g., correcting "Tuesday" to "Thursday" based on context), the system minimises the need for manual editing and supports natural voice-based correction commands.

- **Natural Command Execution:** Users can navigate and format documents using natural speech commands such as "create a bullet list" or "start a new paragraph." *Vocalink* efficiently manages these tasks, providing greater flexibility and control compared to conventional dictation systems.

- **Multilingual Support:** *Vocalink* facilitates seamless language switching, particularly between English and Roman Urdu/Hinglish, allowing users to transcribe in multiple languages without compromising accuracy or needing additional assistance.

- **Accessibility for Neurodivergent Users:** Designed with inclusivity in mind, *Vocalink* reduces cognitive load by automating task management, document formatting, and error correction. This feature is especially beneficial for neurodivergent users, such as those with dyslexia or ADHD, who may find traditional text generation methods challenging.

2.3 Business Opportunity

The market for assistive technologies, particularly in speech recognition and voice-to-text solutions, presents a significant opportunity for *Vocalink* to capture value and meet the growing demand for inclusive and efficient text creation tools.

- **Total Addressable Market (TAM):** The global assistive technology market, including tools catering to various accessibility needs, is set to grow from **$22.98 billion** in **2023** to **$32.25 billion** by **2030**, at a compound annual growth rate (CAGR) of **4.7%**.

- **Serviceable Available Market (SAM):** The speech recognition segment within this market is expected to expand from **$7.14 billion** in **2024** to **$15.87 billion** by 2030, at a **14.24%** CAGR. This presents a large subset of the assistive technology market where *Vocalink* can target its voice-native text editor.

- **Serviceable Obtainable Market (SOM):** The speech-to-text API market, a direct target for *Vocalink*'s offerings, is projected to grow from **$2.4 billion** in **2021** to **$12.1 billion** by **2031**, achieving a CAGR of **17.8%**. This high-growth submarket highlights the increasing reliance on voice technology solutions for diverse applications.

- **Market Share Potential:** With an estimated share capture of **1-2%** of this obtainable market, *Vocalink* could potentially achieve **$76.8 million** in revenue, demonstrating significant business scalability and opportunity within this growing sector.

2.4 Objectives

- **Improve Accessibility:** Provide an intuitive, easy-to-use tool that caters to neurodivergent users (e.g., those with ADHD, dyslexia, or dysgraphia) and individuals with physical limitations that hinder traditional typing methods.

- **Improve Speech-to-Text Accuracy:** Utilise MoE models to dynamically select relevant AI experts, enhancing voice recognition accuracy, reducing WER, and ensuring near-real-time processing of speech inputs.

- **Natural Command Formatting:** Increase efficiency and accessibility by allowing users to format, navigate, and modify documents using intuitive voice commands, eliminating the need for specific keystrokes.

- **Contextual Understanding:** Incorporate contextual awareness to better understand user intent, automatically correct common speech errors, and refine text based on the context of the conversation.

- **Multilingual Support:** Ensure seamless support for multilingual input, particularly for smooth transitions between languages like Roman Urdu and Hinglish, catering to users who frequently switch between languages.

2.5 Project Scope

2.5.1 Deliverables

Vocalink will be developed in four major iterations, each aimed at gradually adding new functionality and improving system performance. These phases will focus on the following key areas:

- **AI Integration:** Implementation of MoE models for text organisation, error correction, and speech-to-text conversion, enabling near-real-time interaction between the editor and the user.
- **Command Execution:** Integration of natural speech commands for document formatting, modification, and navigation, allowing users to interact with documents without relying on a keyboard.
- **Multilingual Capabilities:** Seamless language switching between Hinglish, Roman Urdu, and English to accommodate diverse user groups.
- **Accessibility Features:** Development of features that automate task management, formatting, and error correction to reduce cognitive load, particularly for neurodivergent individuals.
- **Advanced Document Structuring:** In addition to basic transcription, *Vocalink* will support advanced text structuring through voice commands, enabling the creation of complex documents such as essays, reports, and presentations.
- **Privacy and Security:** Implementation of robust user authentication and encrypted cloud storage to ensure secure handling of voice data.

2.5.2 Project Acceptance Criteria

- **Functional Completion:** The voice-native text editor must support all defined features, including near-real-time dictation, error correction, natural command execution, and multilingual support.
- **Accuracy and Performance:** The system must meet predefined accuracy thresholds (e.g., WER less than $\mathbf{5\%}$) and process voice inputs with near-real-time responsiveness.
- **Accessibility and Inclusivity:** The system must meet accessibility standards and demonstrate practical use for neurodivergent individuals and users with physical disabilities. User feedback from targeted test groups will be integral to acceptance.

- **User and Technical Documentation:** Complete, clear, and accurate documentation must be provided for both end users and developers.

- **Stakeholder Sign-Off:** All stakeholders, including product developers and university representatives, must approve the final deliverables.

2.5.3 Exclusions

- **Hardware Requirements:** The project will not include any hardware devices or accessories (e.g., microphones, headsets) for voice input.

- **Unsupported Languages:** Apart from English, Roman Urdu, and Hinglish, no additional languages will be supported in the initial release.

- **Third-Party Integration:** The project will not support any third-party integrations, including with word processing tools such as *Google Docs* and *Microsoft Word*.

- **Mobile Application Development:** Development of mobile versions or applications will not be part of the initial scope of this project.

2.6 Constraints

- **Feature Limitations:** The initial release of *Vocalink* will focus on core functionality, such as near-real-time dictation, voice-based text formatting, and limited natural language processing. Advanced features like integration with third-party tools (e.g., Google Docs or Microsoft Word) or broader language support will be outside the project scope.

- **Platform Focus:** *Vocalink* will be developed primarily for web and desktop platforms. Mobile applications or tablet-specific versions are not included in the current project scope and may be considered for future iterations.

- **Language Support:** Multilingual support in the initial scope will be limited to English, Roman Urdu, and Hinglish. Expanding language support to other languages is beyond the scope of this phase.

- **Limited Third-Party Integration:** The project will not include any integrations with third-party applications, such as word processing tools, beyond the core functionalities of the text editor itself, although cloud storage will be supported.

- **Data Privacy and Security:** While the project will include basic security and privacy measures, advanced data encryption or compliance with specific privacy regulations (e.g., GDPR) is not part of the initial scope. These may be addressed in future iterations.
- **Accessibility Features:** Although inclusivity is a priority, the initial scope will focus on key accessibility features for neurodivergent individuals. Comprehensive support for all global accessibility standards may be deferred for future updates.
- **Time and Budget Constraints:** The project scope is limited by a strict timeline and budget, which restricts the ability to implement non-essential features or extend the functionality beyond what is planned for the initial release.

2.7 Stakeholders Description

The primary stakeholders are the *National University of Computer and Emerging Sciences*, providing academic support, the *Knowledge Discovery and Data Science (KDD) Lab*, where students and faculty develop the project, and the end users, including neurodivergent individuals and professionals, ensuring alignment with educational, technical, and user needs.

2.7.1 Stakeholders Summary

The project involves several key stakeholders, each with distinct goals and challenges that shape its development:

- **End Users (Neurodivergent Individuals and Professionals):** These are the primary beneficiaries of the software. Their feedback and needs drive functional requirements, such as ease of use, hands-free operation, and reducing cognitive load. Their challenges (e.g., accuracy and flexibility of existing tools) are crucial in defining system requirements and constraints.
- **Students (Product Developers and Engineers):** As the developers, we are responsible for implementing the system, adhering to technical specifications, and ensuring that it meets the project's goals. Our challenges (e.g., balancing performance and system scalability) impact design decisions and non-functional requirements.
- **Knowledge Discovery and Data Science (KDD) Lab:** As a research-driven entity, the Knowledge Discovery and Data Science (KDD) Lab facilitates technical research and student-faculty collaboration, emphasising the development of practical and scalable solutions while coordinating efforts among contributors.

- **National University of Computer and Emerging Sciences:** The university serves as a resource provider and ensures the project is aligned with academic goals. Additionally, it helps meet educational objectives by supporting both research and development efforts.

2.7.2 Key High-Level Goals and Problems of Stakeholders

Stakeholder	Key Goals	Key Problems
End Users (Neurodivergent Individuals and Professionals)	• Seek an intuitive, hands-free text creation tool. • Minimise cognitive load and enhance productivity. • Seamless integration into workflows.	• Existing tools lack flexibility, accuracy, and natural interaction. • Traditional dictation tools are inaccurate and command-driven. • Manual post-editing increases frustration and reduces efficiency.
Students (Product Developers and Engineers)	• Build a highly accurate, low-latency voice recognition system. • Reduce error rates while integrating contextual awareness. • Optimise for Roman Urdu and Hinglish support.	• Balancing accuracy with low computational overhead. • Technical hurdles in integrating complex NLP features. • Ensuring scalability and language diversity.
Knowledge Discovery and Data Science (KDD) Lab	• Develop the project with student and faculty collaboration. • Ensure technical alignment with cutting-edge research.	• Coordinating research and development efforts between students and faculty. • Maintaining focus on practical, scalable solutions.
National University of Computer and Emerging Sciences	• Provide academic support and resources for the project. • Ensure alignment with educational goals.	• Balancing academic objectives with the project's technical requirements. • Ensuring long-term sustainability of resources and support.

Table 2.1: Primary Objectives and Challenges of Stakeholders

Chapter 3

Software Requirement Specifications

3.1 List of Features

- **Near-Real-Time Voice-to-Text Conversion:** Convert spoken input into text with minimal latency, allowing for smooth and continuous dictation.
- **Natural Language Commands:** Recognise and execute natural voice commands for text formatting and document navigation, such as "create a new paragraph," "bold this text," or "move to the next section."
- **Multilingual Support:** Seamless support for English, Roman Urdu, and Hinglish, allowing users to dictate and switch between these languages without manual changes.
- **Adaptive Formatting:** Automatically format text based on voice commands, such as creating bullet points, numbered lists, and adjusting paragraph styles.
- **Contextual Error Correction:** Leverage contextual understanding to automatically correct common speech recognition errors (e.g., homophones or mispronunciations) and suggest accurate alternatives.
- **Meeting Notes Transcription:** Allow users to transcribe meeting notes by pressing "Start" to begin and "End" to stop, with the system automatically recording and saving the transcription with timestamps.
- **Speech Playback and Editing:** Provide the option to play back transcribed speech, enabling users to listen to their dictated content and make corrections or revisions via voice commands.
- **Minimal User Interface:** Feature a simple, clutter-free user interface with essential controls, allowing users to focus on dictation and voice commands without unnecessary distractions.

3.2 Functional Requirements

- **Voice-to-Text Conversion:**
  - The system must accurately convert speech into text in near-real-time.
  - Users should be able to dictate continuously without significant latency.
  - The system should support different accents and variations in speech, specifically English, Roman Urdu, and Hinglish.
- **Natural Command Recognition:**
  - The system must recognize and execute natural language commands for formatting and document navigation (e.g., "create a bullet list," "start a new paragraph," "delete the last sentence").
  - Commands should be processed without predefined phrases, allowing for flexibility in user input.
- **Multilingual Support:**
  - The system must allow seamless switching between supported languages (English, Roman Urdu, and Hinglish) within the same document.
  - The system should accurately transcribe speech in multiple languages and maintain the formatting consistency.
- **Meeting Notes Transcription:**
  - The system must allow users to start and stop transcription of meeting notes by pressing "Start" and "End" buttons.
  - The system must automatically format meeting notes with timestamps, speaker identification (if applicable), and key points in bullet or numbered lists.
- **Contextual Understanding:**
  - The system must understand the context of spoken input and automatically correct common errors (e.g., correcting "Tuesday" to "Thursday" if the context suggests a different day).
  - Contextual understanding should enhance dictation accuracy by adapting to the content of the document.
- **Adaptive Error Correction:**
  - The system should automatically identify and correct common speech-to-text errors (e.g., homophones, mispronunciations) without requiring user intervention.

- The system should provide suggestions for corrections when errors cannot be resolved automatically.

- **Document Formatting:**
  - Users should be able to format text using voice commands (e.g., bold, italic, underline, create bullet points, and numbered lists).
  - The system must allow users to navigate through the document using voice commands, such as moving to a specific section or page.

- **Minimal User Interface:**
  - The system must provide a simple, intuitive interface with only essential controls, minimising cognitive load.
  - Users should primarily interact via voice commands, reducing the need for manual input.
  - Advanced options should remain hidden unless explicitly requested, maintaining a clutter-free environment.

- **Cloud Storage Integration:**
  - The system must allow users to save and retrieve documents from cloud storage, ensuring access to files across devices.
  - Seamlessly sync documents across all devices via the cloud.

- **Speech Playback and Editing:**
  - The system must allow users to play back dictated text, providing an option to listen to what has been transcribed.
  - Users should be able to select and edit specific sections of text using voice commands.

- **System Notifications and Feedback:**
  - The system must provide real-time feedback on the status of dictation (e.g., active listening, processing, or errors).
  - Users should receive notifications when errors occur or when corrections are applied automatically.

3.3 Non-Functional Requirements

- **Performance:**
  - The system must provide near-real-time processing of speech inputs with a maximum latency of **5 seconds** between input and text display (dependent on internet connectivity).
  - The system should handle up to **60 minutes** of continuous dictation without significant performance degradation.
  - Voice recognition accuracy should maintain a WER of less than **5%** under optimal conditions.
- **Scalability:**
  - *Vocalink* must be able to scale to support multiple users simultaneously without affecting system performance.
  - The system should efficiently handle large document sizes (e.g., up to **5,000 words**) without a noticeable impact on response time or user experience.
  - Cloud storage integration must be scalable, allowing multiple documents to be stored and retrieved simultaneously across different devices.
- **Usability:**
  - The user interface must be minimal and intuitive, with a clear and easy-to-navigate layout, ensuring that users with minimal technical experience can operate the system without extensive training.
  - The system must respond to user commands (e.g., voice commands, button presses) within 1 second to provide a smooth and responsive user experience.
- **Security:**
  - All user data, including voice inputs and transcriptions, must be encrypted both in transit and at rest, adhering to industry-standard encryption protocols (e.g., *AES-256*).
  - User authentication must be secure, supporting *OAuth 2.0* for single sign-on (SSO) and multi-factor authentication (MFA) to prevent unauthorised access.
- **Privacy:**
  - The system must ensure that all voice data is processed securely and deleted after transcription, unless explicitly saved by the user.

  - Users should have full control over their data, with the ability to delete stored transcriptions and associated metadata at any time.

- **Maintainability:**
  - The system should be modular, allowing for easy updates and maintenance of individual components (e.g., voice recognition, UI) without impacting the entire system.
  - System documentation must be comprehensive and up to date, facilitating future maintenance and feature enhancements by developers.

3.4 Use Case

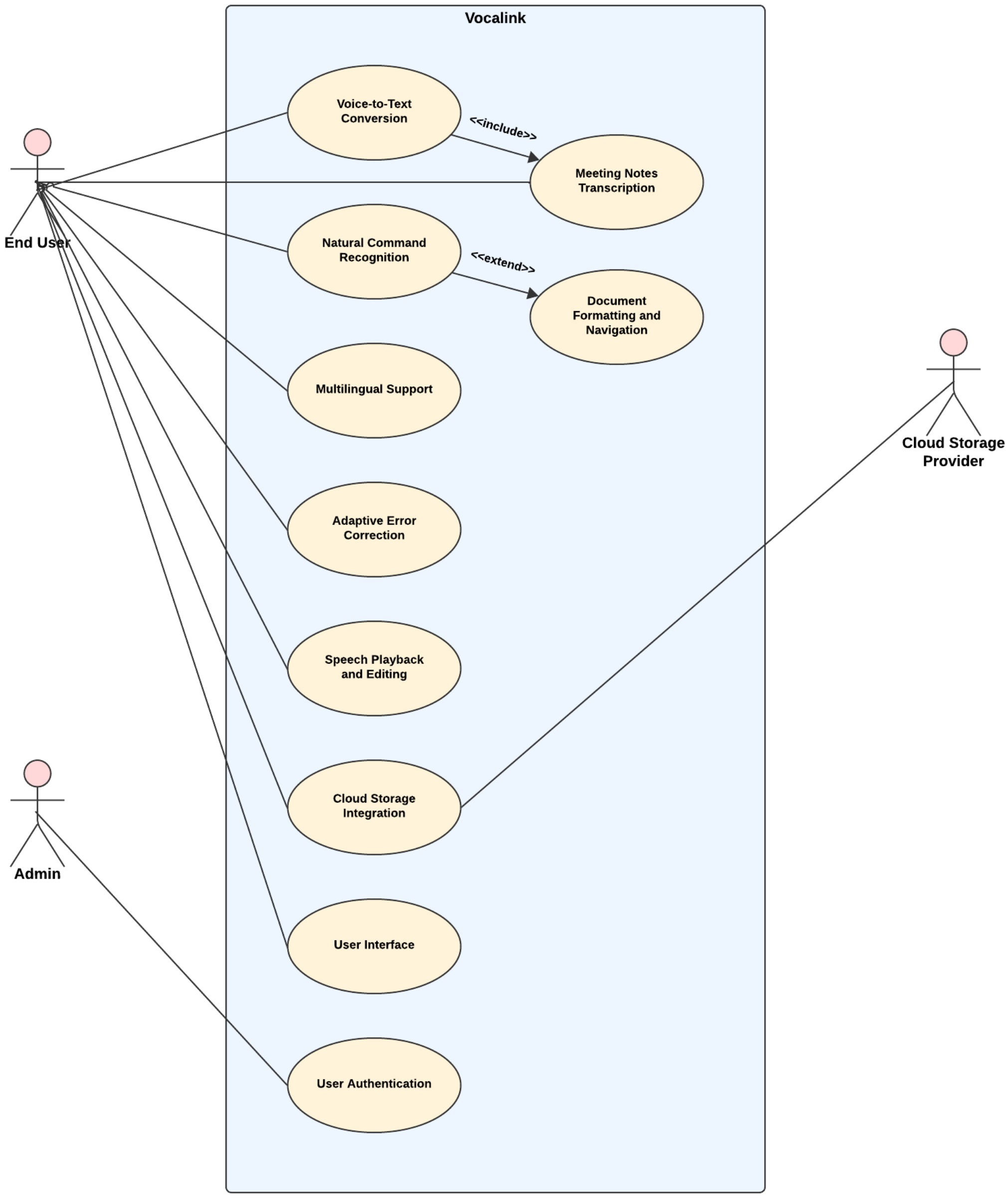

Figure 3.1: Use Case Diagram of Admin, End User, and System Interactions

Use Case ID	UC-01
Use Case Name	Voice-to-Text Conversion
Description	Converts the user's spoken input into text in near-real-time.
Primary Actor	End User
Preconditions	• The application is open. • The microphone is functional.
Postconditions	• Spoken input is transcribed into text in the document editor.
Main Flow	1. User activates the "Start" button to begin dictation. 2. The system listens for speech input. 3. The system converts speech to text and displays it in the document. 4. User presses "Stop" to end dictation.
Alternate Flows	• If the microphone is not working, the system shows an error message. • If background noise is detected, the system notifies the user and pauses transcription.

Table 3.1: Use Case UC-01: Voice-to-Text Conversion

Use Case ID	UC-02
Use Case Name	Natural Command Recognition
Description	Executes voice commands for document formatting and navigation.
Primary Actor	End User
Preconditions	• The system is actively listening. • A document is open.
Postconditions	• The voice command is executed (e.g., create a new paragraph, format text).
Main Flow	1. User issues a voice command (e.g., "create a new paragraph"). 2. The system processes the command and executes the requested action. 3. The system provides real-time feedback on the action completion.
Alternate Flows	• If the command is not recognised, the system prompts the user to try again. • If there are multiple actions for a command, the system asks for clarification.

Table 3.2: Use Case UC-02: Natural Command Recognition

Use Case ID	UC-03
Use Case Name	Multilingual Support
Description	Allows users to switch between languages (English, Roman Urdu, Hinglish) during dictation.
Primary Actor	End User
Preconditions	• The user is dictating or issuing commands.
Postconditions	• The text is transcribed in the correct language based on the input.
Main Flow	1. User starts dictating in one language. 2. User switches to another language (e.g., from English to Roman Urdu). 3. The system detects the language switch and continues transcription in the new language.
Alternate Flows	• If the system doesn't detect the language switch, the user manually selects the new language.

Table 3.3: Use Case UC-03: Multilingual Support

Use Case ID	UC-04
Use Case Name	Adaptive Error Correction
Description	Automatically detects and corrects common speech-to-text errors in real-time.
Primary Actor	End User
Preconditions	• The user is dictating, and the system is actively transcribing.
Postconditions	• The text is transcribed with errors automatically corrected based on context.
Main Flow	1. The user dictates a sentence. 2. The system detects an error in the transcription. 3. The system automatically corrects the error based on context (e.g., correcting "their" to "there"). 4. The corrected text is displayed to the user.
Alternate Flows	• If the system cannot correct the error, it highlights the text and prompts the user for manual correction.

Table 3.4: Use Case UC-04: Adaptive Error Correction

Use Case ID	UC-05
Use Case Name	Speech Playback and Editing
Description	Allows users to play back transcribed speech and make corrections or edits using voice commands.
Primary Actor	End User
Preconditions	• A document with transcribed text is open. • The system is in active listening mode for voice commands.
Postconditions	• The user can listen to the transcribed speech and make corrections or edits.
Main Flow	1. The user selects a section of the document to play back. 2. The system plays back the transcribed speech. 3. The user issues voice commands to correct or edit the text (e.g., "replace 'Tuesday' with 'Thursday'"). 4. The system updates the text accordingly and confirms the change.
Alternate Flows	• If the system cannot identify the text to edit, it prompts the user to clarify the command. • If the playback fails, the system alerts the user and provides troubleshooting steps.

Table 3.5: Use Case UC-05: Speech Playback and Editing

Use Case ID	UC-06
Use Case Name	Cloud Storage Integration
Description	Saves and retrieves user documents from the cloud for access across devices.
Primary Actor	End User
Preconditions	• The user is logged in, and the system has access to cloud storage.
Postconditions	• The document is saved to or retrieved from cloud storage.
Main Flow	1. User selects the option to save or open a document. 2. The system syncs the document with cloud storage. 3. The document is saved to the cloud or retrieved and opened for editing.
Alternate Flows	• If cloud storage is unavailable, the system notifies the user and saves the document locally until syncing is possible.

Table 3.6: Use Case UC-06: Cloud Storage Integration

Use Case ID	UC-07
Use Case Name	User Interface
Description	Provides a minimal and intuitive interface that allows users to interact with the system primarily through voice commands, minimising manual interaction.
Primary Actor	End User
Preconditions	• The system is launched and ready for interaction. • The user has an active session and is logged in.
Postconditions	• The user can access the core system features via a minimal user interface with voice or manual commands.
Main Flow	1. User launches the application and views the minimal interface. 2. The system displays essential controls, such as "Start Dictation," "Stop Dictation," and document management. 3. The user issues voice commands or interacts with basic controls to navigate the interface and perform tasks. 4. The system responds to the user's input, executing commands and updating the interface.
Alternate Flows	• If the user requests advanced options, the system expands the interface to show additional features. • If the system encounters an issue with voice commands, the user is prompted to use manual controls.

Table 3.7: Use Case UC-07: User Interface

Use Case ID	UC-08
Use Case Name	User Authentication
Description	Admin authenticates users and manages user access by controlling authentication settings through the Supabase dashboard.
Primary Actor	Admin
Preconditions	• The system is active and ready for user login. • The admin has valid credentials and access to the Supabase dashboard.
Postconditions	• The admin successfully logs into the Supabase dashboard and can manage user authentication settings (e.g., granting or revoking access). • If authentication fails, access to the Supabase dashboard is denied.
Main Flow	1. Admin navigates to the Supabase dashboard. 2. Admin enters their login credentials (username and password). 3. The system validates the credentials against the stored user database. 4. If credentials are valid, the admin is logged in and gains access to the Supabase dashboard. 5. The admin can manage user roles and permissions, including granting or revoking access, through the Supabase authentication settings.
Alternate Flows	• If the credentials are invalid, the system displays an error message, and the admin is prompted to retry login. • If the system detects suspicious login attempts, the admin is locked out temporarily, and a security notification is triggered.

Table 3.8: Use Case UC-08: User Authentication

Use Case ID	UC-09
Use Case Name	Meeting Notes Transcription
Description	Transcribes meeting notes when the user presses "Start" and stops transcribing when the user presses "End."
Primary Actor	End User
Preconditions	• The user has initiated a meeting transcription, and the microphone is functional.
Postconditions	• The meeting notes are saved with timestamps indicating the start and end times.
Main Flow	1. User presses the "Start" button to begin transcription. 2. The system transcribes spoken content in near-real-time. 3. User presses "End" to stop transcription. 4. The system saves the notes with timestamps indicating start and end.
Alternate Flows	• If the microphone stops working, the system alerts the user and pauses transcription.

Table 3.9: Use Case UC-09: Meeting Notes Transcription

Use Case ID	UC-10
Use Case Name	Document Formatting and Navigation
Description	Allows users to format documents (e.g., bullet points, bold text) and navigate through sections using voice commands or minimal manual input.
Primary Actor	End User
Preconditions	• A document is open for editing. • The system is actively listening for voice commands or ready for manual input.
Postconditions	• The document is formatted or navigated as per the user's input.
Main Flow	1. User issues a voice command to format the document (e.g., "create a bullet list"). 2. The system applies the requested formatting (e.g., bullet points are inserted). 3. User issues a navigation command (e.g., "go to the last section"). 4. The system moves to the specified section of the document.
Alternate Flows	• If the system does not recognise a command, it prompts the user to repeat or clarify the request. • If multiple sections match the user's navigation command, the system asks for further clarification.

Table 3.10: Use Case UC-10: Document Formatting and Navigation

3.5 Iteration Plan

The development of *Vocalink* will follow a structured iteration plan to ensure incremental progress and refinement of the system. Each iteration will focus on specific development goals, testing, and feature integration, laying the foundation for the final product. The following sections outline the key objectives and activities for each iteration.

3.5.1 Iteration 1 (Q1/Fall-2024): Initial Development Phase

3.5.1.1 Goals

- Establish the core functionality of *Vocalink* by integrating essential components, including Speech-to-Text (STT) and Automatic Speech Recognition (ASR) capabilities.
- Develop a working prototype that demonstrates the basic features and serves as a foundation for future iterations.

3.5.1.2 Key Activities

- **STT/ASR Integration:** Integrate STT and ASR systems to enable basic transcription functionality within the application.
- **Voice Activity Detection (VAD) Integration:** Implement Voice Activity Detection (VAD) to accurately detect when the user is speaking, ensuring smooth transitions between voice input and silence.
- **Prototype Development:** Build an initial prototype of the system, allowing for basic user interactions such as voice input and text transcription.
- **Integration of MoE Model:** Begin the integration of the MoE model to improve transcription accuracy and contextual understanding in the voice recognition system.

3.5.1.3 Deliverables

- Basic working prototype with functional STT, ASR, and VAD features.
- Initial implementation of the MoE model for enhanced accuracy.

3.5.2 Iteration 2 (Q2/Fall-2024): Advanced Feature Development and Testing

3.5.2.1 Goals

- Build upon the initial prototype by adding advanced features and improving system performance through rigorous testing and refinement.
- Develop the first full application prototype with additional user-facing features.

3.5.2.2 Key Activities

- **Preliminary Testing Phase:** Conduct initial tests to evaluate the performance and accuracy of the integrated STT/ASR systems and the MoE model. Identify any issues or areas for improvement.
- **Development of Advanced Features:** Begin implementing advanced features such as natural language commands for document formatting, multilingual support, and error correction.
- **Initial Application Prototype:** Create a more refined application prototype that includes the core functionality and advanced features, providing a more complete user experience.

3.5.2.3 Deliverables

- Preliminary test reports on STT/ASR accuracy and performance.
- Initial application prototype with basic and advanced features (e.g., formatting commands, multilingual support).

3.5.3 Iteration 3 (Q1/Spring-2025): Security, Deployment, and Final Testing

3.5.3.1 Goals

- Implement robust security measures, deploy the system to the cloud, and gather user feedback.
- Ensure the system is secure, scalable, and ready for broader testing and evaluation.

3.5.3.2 Key Activities

- **Security Integration (Guardrails):** Implement security features to protect user data and ensure compliance with security standards. This will include guardrails and security protocols for both the web and mobile platforms.

- **Cloud Deployment (Web/Mobile):** Deploy the system to the cloud, ensuring that it is accessible from both web and mobile platforms. This will involve setting up infrastructure for scalability and performance.

- **User Feedback Collection:** Initiate user feedback sessions to gather insights on the system's usability, performance, and features.

- **Final Testing:** Conduct final rounds of testing across various platforms to ensure system stability, scalability, and performance.

3.5.3.3 Deliverables

- Full security implementation, including guardrails.

- Deployed system accessible via web and mobile platforms.

- User feedback reports for future improvements.

- Final testing reports, ensuring readiness for the next phase.

3.5.4 Iteration 4 (Q2/Spring-2025): Performance Optimisation and Launch Preparations

3.5.4.1 Goals

- Finalise the product through performance optimisation, refining existing features, and preparing for the official launch.

- Continue to enhance the user interface and experience based on feedback.

3.5.4.2 Key Activities

- **Performance Optimisation:** Refine the system's performance to ensure fast, responsive interactions, focusing on areas such as transcription speed, accuracy, and scalability.

- **Improving Existing Features:** Make improvements to existing features, such as voice commands, document formatting, and multilingual support, based on testing and user feedback.
- **Finalising Launch Details:** Prepare for the public launch by completing all required documentation and launch-related tasks.
- **Iterative UI/UX Enhancements:** Continuously improve the user interface and experience based on iterative testing and feedback, ensuring an intuitive and user-friendly design for all platforms.

3.5.4.3 Deliverables

- Optimised system performance, ready for production.
- Finalised feature set with improvements based on feedback.
- Complete launch plan and documentation.
- Refined UI/UX design across all platforms.

Chapter 4

Software Design

The software design of *Vocalink* is centered around a modular and scalable architecture that leverages the MoE model to provide high-accuracy, low-latency voice-to-text conversion and natural language processing capabilities. The design emphasises procedural flow, ensuring that the system processes input data in a streamlined and efficient manner.

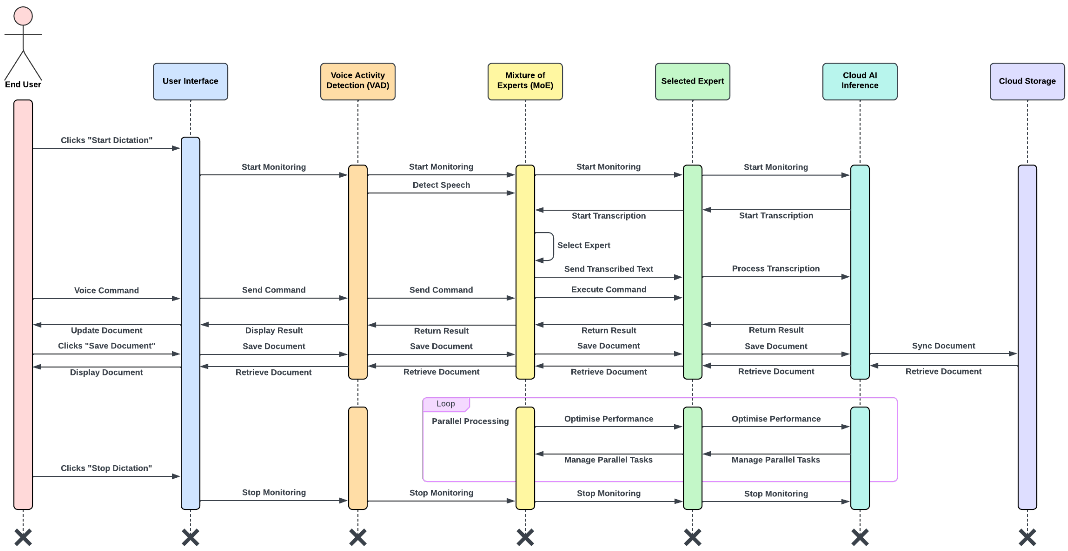

Figure 4.1: Sequence Diagram of Vocalink's System Design

4.1 Modular Architecture

The architecture of *Vocalink* is divided into several key modules, each responsible for a specific set of tasks. This modular approach ensures that each component can be developed, tested, and optimised independently while still functioning cohesively as part of the overall system. The primary modules include:

4.1.1 User Interface

- The user interacts with the system through this interface, issuing voice commands or managing documents. The interface is minimal and intuitive, ensuring ease of use.
- The user interface connects directly with the speech processing module to begin voice input and transcription.

4.1.2 Speech Processing

This module handles all audio processing, from detecting the user's speech to converting it into text. The key components include:

- **Voice Activity Detection (VAD):** Detects when the user begins speaking and when they stop, ensuring efficient speech processing.
- **Automatic Speech Recognition (ASR):** Converts the detected voice input into text.
- **Speech-to-Text (STT):** Finalises the transcription of speech into text that will be displayed in the user interface.

4.1.3 Mixture of Experts (MoE) Model

The heart of the system's intelligence lies in the MoE model. It consists of a gating network (see Section 5.1.1 for details) that dynamically selects which experts to activate based on the user's input. The experts include specialised submodules focused on:

- **STT Accuracy:** Enhances the accuracy of the speech-to-text transcription.
- **Contextual Understanding:** Ensures the system comprehends the context behind the spoken words, improving the accuracy of transcriptions.

- **Natural Language Command Recognition:** Processes commands embedded in speech (e.g., “bold this word,” “create a new paragraph”) and forwards them to the application logic.

4.1.4 Application Logic

- **Error Correction Module:** Automatically corrects common speech-to-text errors based on context, improving the overall accuracy of the transcribed document.
- **Command Processing Engine:** Executes recognized commands for document formatting, such as bolding text or navigating between sections.

4.1.5 Data Storage

The system integrates cloud storage to ensure all user profiles and documents are securely stored and easily accessible across devices.

- **User Profiles:** Stores user-specific data, such as preferences, access permissions, and usage history.
- **Document Storage:** Handles the saving and retrieval of documents that users interact with. All documents are synced to the cloud for access from multiple platforms.

4.1.6 Security

To ensure data safety, the system incorporates several security measures:

- **Authentication and Authorisation:** Verifies the identity of users and ensures they have the correct permissions to access the system.
- **Data Encryption:** Ensures all data (both in transit and at rest) is encrypted to protect sensitive information.

4.1.7 Integration

- **Cloud Deployment:** The entire system is deployed in the cloud, making it accessible from anywhere and ensuring scalability.
- **Cloud AI Inference:** Utilises cloud-based AI inference (see Section 5.2.1 for details) to process voice commands and execute transcription and command recognition with minimal latency.

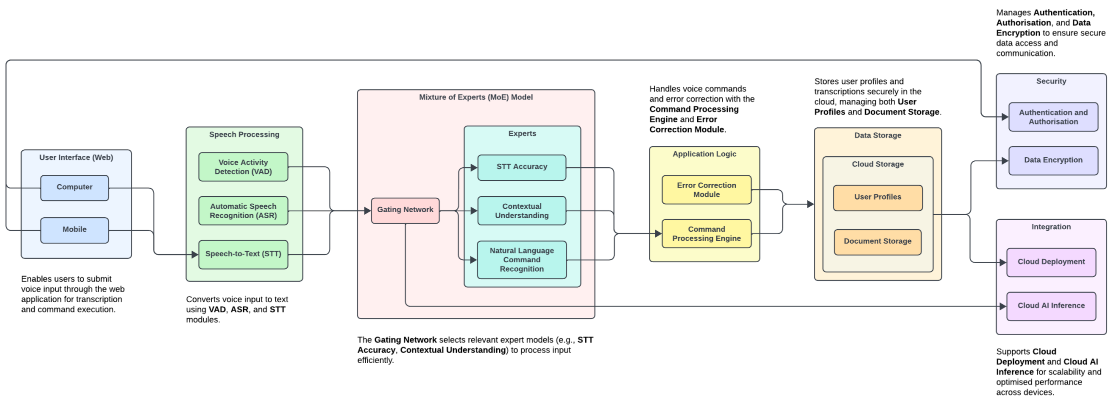

Figure 4.2: System Architecture High-Level Overview

4.2 Procedural Flow

Vocalink uses a procedural flow model, where each task is executed in a step-by-step sequence, ensuring that the system processes inputs efficiently. The following outlines the typical procedural flow of the system:

4.2.1 User Input Initiation

- The user begins an interaction by accessing the interface on a web-based platform, either on a computer or mobile device. They press a button to start voice recording or use a voice command to begin.
- The VAD module in the speech processing system detects the user's voice, initiating the transcription process.

4.2.2 Speech Processing

- Once voice input is detected, the ASR module begins converting the audio signals into a structured text format.
- The transcribed text is generated using the STT system, which relies on the MoE model for improved accuracy. The text is passed on to the gating network within the MoE.

4.2.3 Dynamic Expert Selection

The gating network selects the most appropriate experts to handle the incoming voice data. Based on the input, different experts are activated:

- **STT Accuracy Expert:** Ensures the transcription is as accurate as possible, especially for complex or nuanced speech.
- **Contextual Understanding Expert:** Assesses the context of the input to improve the accuracy of the transcribed text in relation to its meaning and structure.
- **Natural Language Command Recognition Expert:** Identifies specific commands within the input (e.g., "bold this," "create a bullet list") and forwards them to the application logic.

4.2.4 Command Processing and Error Correction

Once the transcription and command recognition are complete, the system enters the application logic phase.

- The command processing engine interprets the voice commands and executes them within the document. For instance, if the user said, "bold this word," the command is processed, and the formatting is applied.

  Simultaneously, the error correction module automatically identifies and corrects any common errors in the transcription (e.g., homophones, misrecognised words), improving the quality of the final output.

4.2.5 Document Update

- After processing the input, the document is updated with the transcribed text and any associated formatting or command-driven changes (e.g., text formatting, navigation changes).
- The updates are shown in the User Interface, providing the user with real-time feedback on their input.

4.2.6 Cloud Storage and Data Handling

Once the document is updated, it is saved to the cloud storage module. This includes:

- **User Profiles:** The system checks the user profile to ensure personalised settings (e.g., document preferences) are applied.

  Document Storage: The document is automatically saved and synced in the cloud, ensuring access across multiple devices.

The cloud storage also facilitates version control and document sharing across platforms, allowing users to retrieve documents seamlessly from either a desktop or mobile interface.

4.2.7 Real-Time Feedback and Notifications

- Throughout the interaction, the system continuously provides feedback to the user via the user interface. Notifications such as "Listening," "Processing," and "Ready" are displayed to indicate the current status.

- If an error occurs (e.g., speech recognition failure, unclear commands), the system alerts the user to try again or clarifies the command input.

4.2.8 Continuous Integration and Cloud AI Inference

- As part of the system's scalability and efficiency, the cloud AI inference module is used to continuously process user inputs with high speed and minimal latency. This ensures real-time interaction, even with complex commands or large-scale inputs.
- The cloud deployment ensures that updates to the software or improvements in the MoE model are automatically integrated without disrupting the user experience.

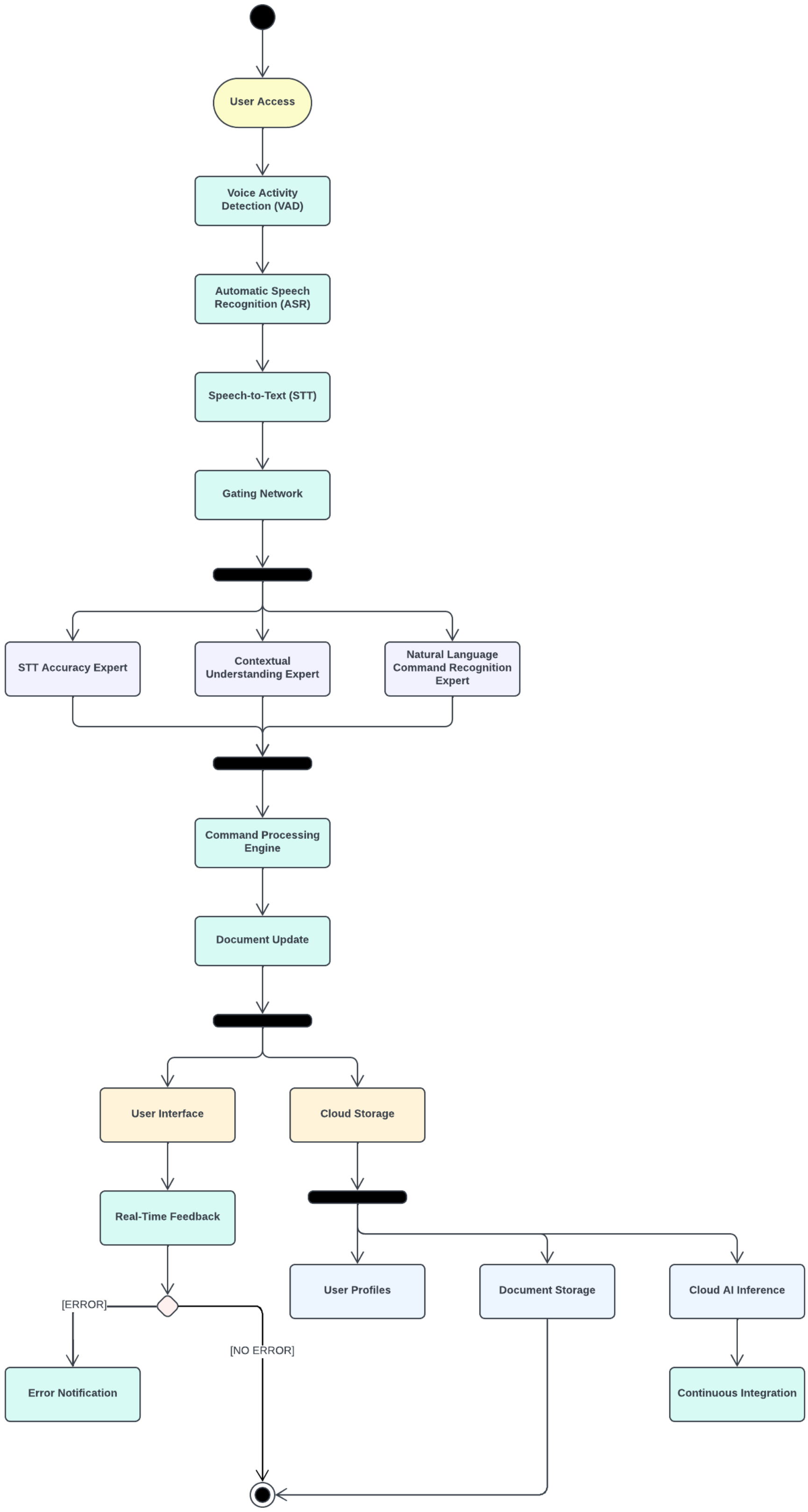

Figure 4.3: Procedural Flow Activity Diagram for System Design

4.3 Data Flow

The data flow in *Vocalink* begins with the user's voice input, leading to document generation, formatting, and storage. Voice input is captured, pre-processed, and sent to the STT engine, with the MoE model dynamically selecting experts to ensure high transcription accuracy.

The resulting text moves through the document formatting and navigation module, which interprets natural language commands for structuring (e.g., bullet points or section navigation). Once formatted, the document is saved locally or to cloud storage for future access and editing. User authentication and access control ensure that only authorised users can interact with the system.

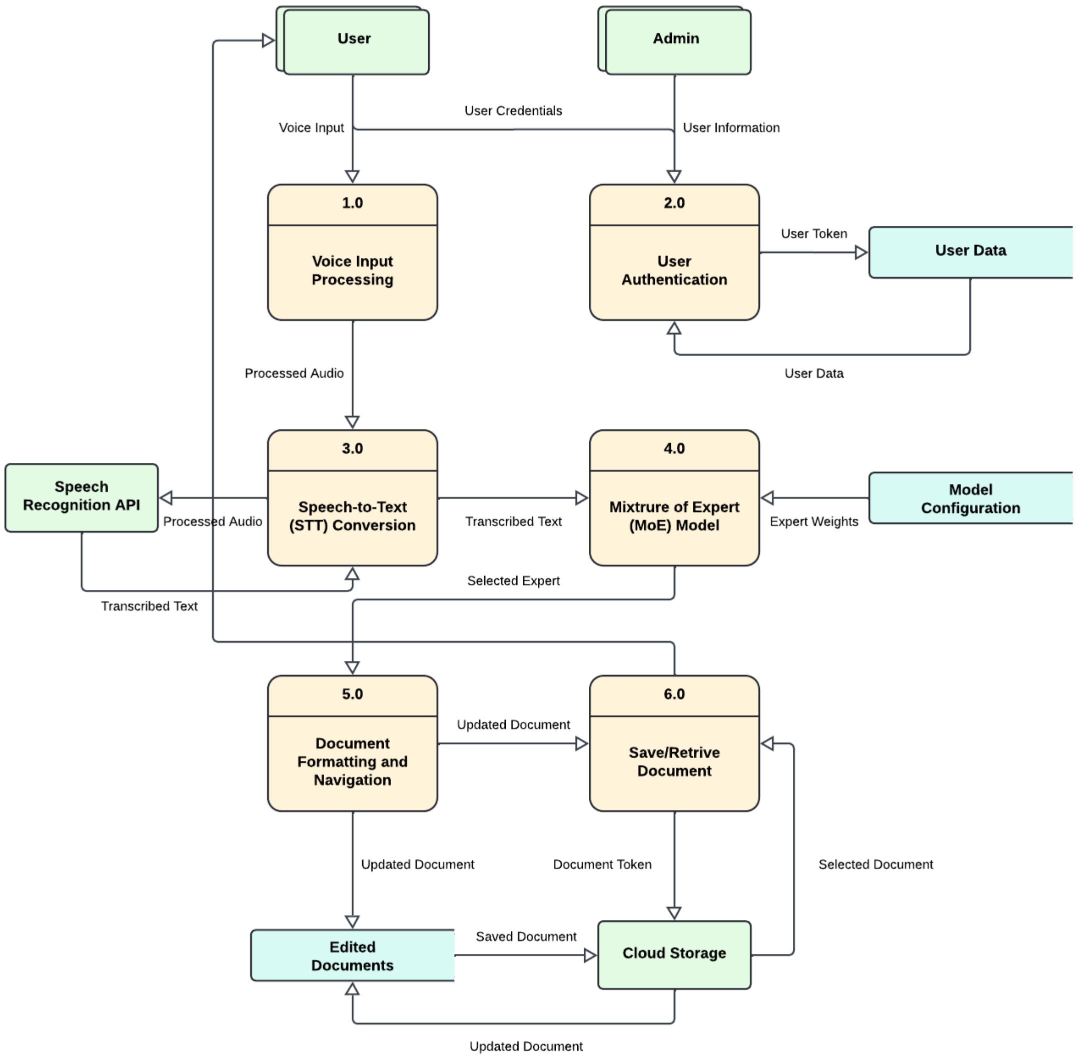

Figure 4.4: Data Flow Diagram for Input and Output Handling

The following schema provides a conceptual representation of the data structure aligned with the outlined data flow:

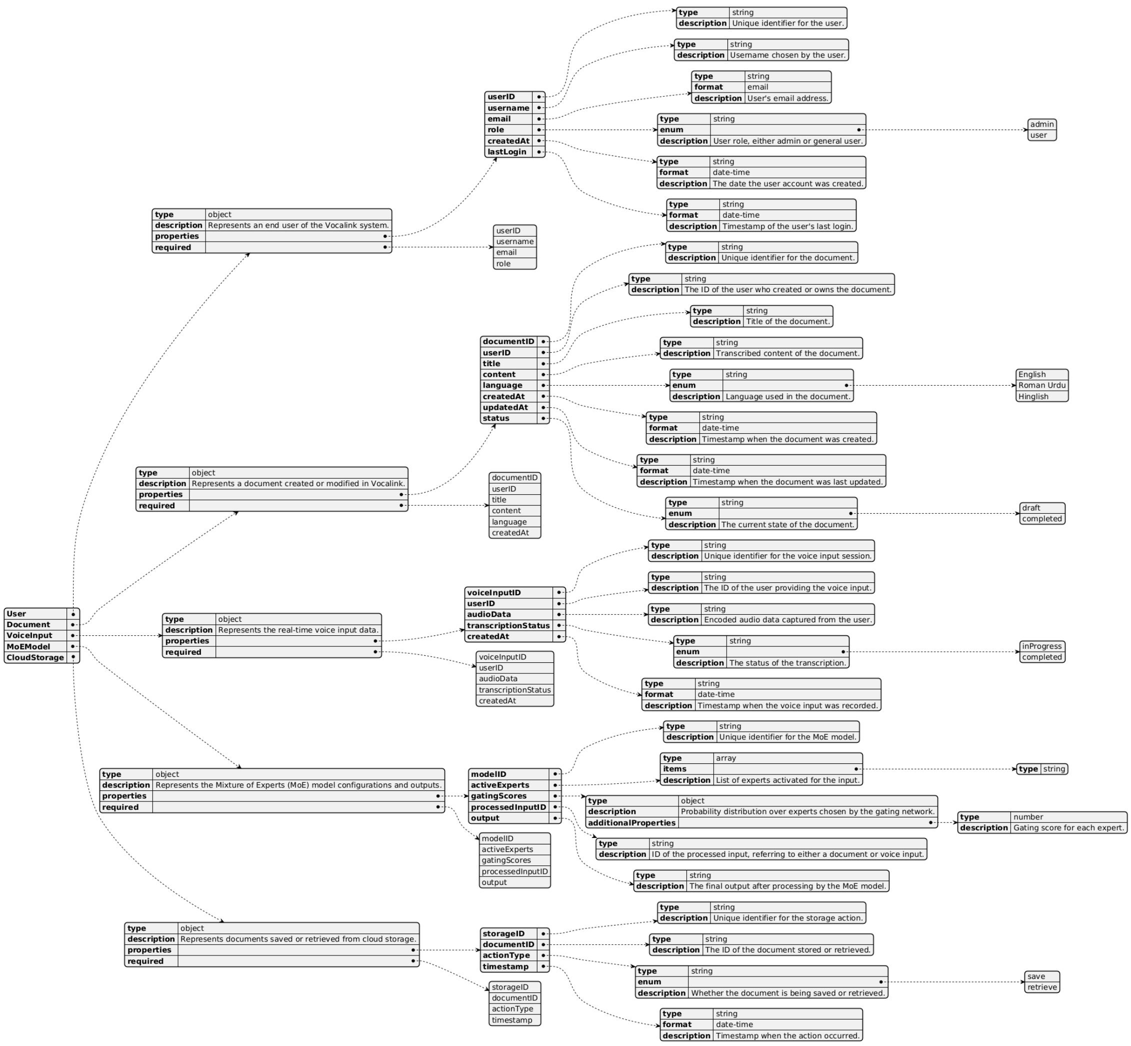

Figure 4.5: Data Representation Diagram in JavaScript Object Notation (JSON) Schema Format

Chapter 5

Implementation

The implementation of *Vocalink* leverages the MoE architecture to optimise voice-to-text conversion, natural command recognition, and other advanced features. The MoE model is an advanced neural network approach designed to enhance both efficiency and performance by dynamically activating specific experts tailored to handle different input tasks. In this section, we outline the mathematical models and key components of the implementation.

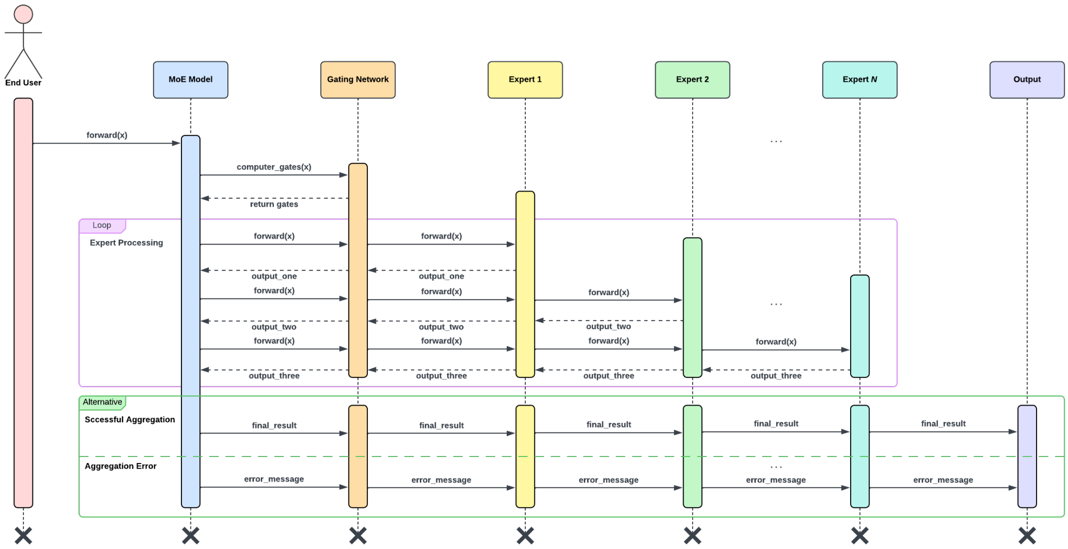

Figure 5.1: Sequence Diagram for Mixture of Experts (MoE) Architecture

5.1 Mathematical Model

The MoE architecture allows *Vocalink* to dynamically select which experts (sub-networks) should process a given input. This improves system performance by avoiding the need to activate all experts for every task. Instead, the gating mechanism selects only the most relevant experts for processing the input.

Let x represent the input vector, and let $E_1, E_2, \ldots, E_N$ represent the N experts available in the system. The goal is to produce an output y based on the input, using only a subset of the experts for efficient processing. The selection of experts is handled by a gating network, which produces a probability distribution over the experts.

The MoE model can be mathematically expressed as:

$$y = \sum_{i=1}^{N} g_i(x) E_i(x)$$

where:

- $g_i(x)$ is the gating score for expert E_i, representing the probability assigned by the gating network for that expert to process input x.
- $E_i(x)$ is the output of expert E_i when given input x.
- y is the final output, which is a weighted combination of the outputs from the selected experts.

5.1.1 Gating Network

The gating network is a critical component that decides which experts should be activated for a given input. It uses a probability distribution over the experts and is designed to minimise computational overhead by only activating the most relevant experts. The gating mechanism uses the Switched Gated Linear Unit (SwiGLU) activation function, which is defined as:

$$\text{SwiGLU}(\mathbf{x}) = \mathbf{x} \cdot \sigma(W\mathbf{x} + b)$$

where:

- x is the input to the gating network.
- W and b are learned parameters (weights and biases).

- σ is the sigmoid function, which outputs values between 0 and 1.

The SwiGLU activation function enhances the gating network's ability to capture non-linear relationships, ensuring that only the most relevant experts are selected based on the input.

5.1.2 Key Optimisations

Several optimisations have been integrated into the MoE model to ensure that the system operates efficiently:

5.1.2.1 Gate Skipping

This optimisation allows certain inputs to bypass the gating mechanism when they can be directly mapped to a specific expert based on predefined conditions. This reduces the need for gating decisions, improving the overall processing speed.

$$\text{Expert}_{\text{direct}} = E_k(\mathbf{x}) \quad \text{if}\, \mathbf{x} \in C_k$$

where C_k is a predefined condition for expert E_k.

5.1.2.2 Dynamic Routing

Dynamic routing reduces the number of active experts by selecting only a subset based on the input characteristics. This approach minimises computational cost while maintaining performance.

$$y = \sum_{i \in A} g_i(x) E_i(x)$$

where A is the set of active experts selected by the gating mechanism.

5.1.2.3 Weight Sharing

Experts share certain network parameters, reducing memory consumption while maintaining high performance. This is expressed as:

$$E_i(\mathbf{x}) = f_{\text{shared}}(\mathbf{x}) + f^i_{\text{task-specific}}(\mathbf{x})$$

where f_{shared} represents shared parameters, and $f^i_{\text{task-specific}}$ represents task-specific processing.

5.1.2.4 Sparse Activation

Only a small number of experts are activated during each forward pass. This sparse activation pattern reduces the overall computational cost by activating only the most relevant experts for a given input.

$$g(x) \approx \text{Sparse}(g(x))$$

where $\text{Sparse}(g(x))$ represents the gating function outputting mostly zero values for non-relevant experts.

For example, the gating function may produce mostly zeros for several experts, leading to a sparse activation pattern:

$$g_i(\mathbf{x}) = 0 \quad \text{for most } i$$

5.2 Expert Parallelism

Experts are highly effective in high-throughput scenarios involving many machines. Early MoE work introduced MoE layers as a branching design, which caused slowdowns since GPUs aren't optimised for such operations, and network bandwidth became a bottleneck due to communication between devices. This highlights the need for a parallelism approach to make MoE pretraining and inference more efficient.

With expert parallelism, experts are distributed across different workers, each processing a distinct batch of training samples. In non-MoE layers, expert parallelism functions like data parallelism, while for MoE layers, tokens are routed to workers hosting the required experts.

- **Data Parallelism:** The same model weights are copied across all cores, with the data split among the cores.
- **Model Parallelism:** The model is split across cores, while the data is replicated across all cores.
- **Model and Data Parallelism:** Both the model and data are partitioned across cores, with each core processing different data batches.
- **Expert Parallelism:** Experts are distributed across different workers. When combined with data parallelism, each core hosts a different expert, and the data is divided across all cores.

5.2.1 Cloud AI Inference Integration

MoE systems can utilise cloud AI inference platforms, such as *GroqCloud*, to boost the model's efficiency and scalability. These platforms provide APIs that support parallel processing across multiple experts, optimising resource allocation and ensuring low-latency inference.

- **Parallel Processing:** Cloud AI inference platforms like *GroqCloud* enable multiple experts to process inputs concurrently, utilising high-performance hardware to deliver real-time performance.
- **Scalability:** The system can easily scale by adding more experts without performance degradation, due to the efficient resource management inherent to cloud inference platforms like *GroqCloud*.

5.3 Application

The MoE architecture is used to optimise performance, accuracy, and scalability in *Vocalink*, enhancing the following key features:

- **Speech-to-Text (STT) Accuracy:** By dynamically selecting the most relevant experts based on voice input characteristics, the system improves the accuracy of speech recognition, particularly in multilingual contexts.
- **Contextual Understanding:** The gating mechanism ensures that inputs are routed to experts that can process contextual nuances, improving overall comprehension and transcription quality.
- **Natural Command Recognition:** The system effectively identifies and processes natural language commands by routing inputs to specialised experts that handle specific tasks, such as document formatting or navigation.

Chapter 6

Conclusions and Future Work

The project seeks to create a voice-native document creation system leveraging the MoE architecture to enhance transcription accuracy and natural language command recognition. Although the initial focus will be on paraphrasing and Roman Urdu/Hinglish translation, there are several anticipated challenges and opportunities for future growth.

- **Expansion of Task Domains:** Initially, the system will handle a limited set of tasks, but future development can expand to include sentiment analysis, machine translation, and document summarisation. This expansion would require fine-tuning new experts on diverse datasets.
- **Collaborative Expert Processing:** As the range of tasks grows, future versions could enable multiple experts to collaborate dynamically, improving accuracy for more complex tasks by sharing partial results.
- **Data and Training Scalability:** The system's effectiveness relies on high-quality, diverse training data. Ensuring scalability for tasks like Roman Urdu/Hinglish translation will require continuous improvements in dataset quality and coverage.

Vocalink will begin with a solid focus on voice-driven tasks, but its true potential can be unlocked through future expansions. By tackling challenges such as task diversity, expert collaboration, and data scalability, the system could grow into a robust tool, capable of handling more complex voice interactions and supporting a broader range of use cases.

Chapter 7

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